

Quiz (Habanero)

- Q1.** In a two-way table, what does a cell represent?
(A) Total frequency
(B) Relative frequency
(C) Joint frequency
(D) Marginal frequency
(E) Conditional frequency
- Q2.** A two-way table shows the relationship between Temperature and Rainfall. If the table has 3 rows and 4 columns, how many cells are there?
(A) 10
(B) 12
(C) 6
(D) 8
(E) 7
- Q3.** What type of frequency is calculated by dividing a cell frequency by the total frequency?
(A) Joint frequency
(B) Marginal frequency
(C) Relative frequency
(D) Conditional frequency
(E) Total frequency
- Q4.** In a study of the relationship between Ecosystem and Species, which type of table would be used?
(A) One-way table
(B) Two-way table
(C) Bar graph
(D) Histogram
(E) Scatter plot
- Q5.** If a two-way table has 2 rows and 3 columns, and the total frequency is 24, what is the relative frequency of the cell in the first row and first column if its joint frequency is 6?
- (A) 0.25
(B) 0.5
(C) 0.1
(D) 0.3
(E) 0.2
- Q6.** A two-way table shows the relationship between Weather and Tourist Activity. If the relative frequency of sunny weather is 0.4 and the relative frequency of tourist activity is 0.6, what is the relative frequency of sunny weather and tourist activity together if they are independent?
(A) 0.24
(B) 0.4
(C) 0.6
(D) 0.12
(E) 0.36
- Q7.** In a two-way table, what does the marginal frequency of a row represent?
(A) The total frequency of the row
(B) The relative frequency of the row
(C) The joint frequency of the row
(D) The conditional frequency of the row
(E) The total frequency of the table
- Q8.** A two-way table shows the relationship between Temperature Change and Ecosystem. If there is a strong positive association between the two variables, what can be concluded?
(A) As temperature change increases, ecosystem decreases
(B) As temperature change increases, ecosystem increases
(C) As temperature change decreases, ecosystem decreases
(D) As temperature change decreases, ecosystem increases
(E) There is no relationship between temperature change and ecosystem

Open Ended Questions (FRQ)

- Q9.** A two-way table shows the relationship between Rainfall and Crop Yield. The table is as follows:

	Low	Medium	High
Low	10	15	20
Medium	20	30	10
High	30	20	15

Calculate the relative frequency of low rainfall and low crop yield. Show your work and give your answer as a decimal value.

- Q10.** A study examines the relationship between Weather Conditions and Air Quality. The two-way table is as follows:

	Good	Fair
Sunny	25	30
Cloudy	30	20
Rainy	10	25

Analyze the association between weather conditions and air quality. Provide a conclusion based on the data and give at least two pieces of evidence from the table to support your conclusion.

Chef's Tips

- Tip 1: Make sure to read each question carefully and understand what is being asked.
- Tip 2: Use the data in the two-way tables to calculate relative frequencies and analyze associations between variables.
- Tip 3: Show all work and provide clear explanations for free-response questions.